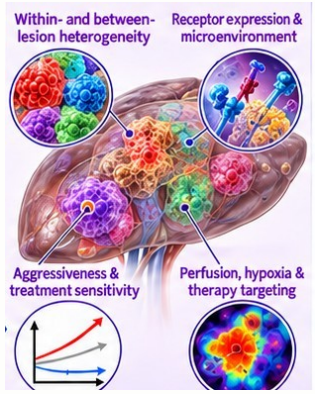
AIM I

Quantitative molecular imaging biomarkers



1 Measure disease

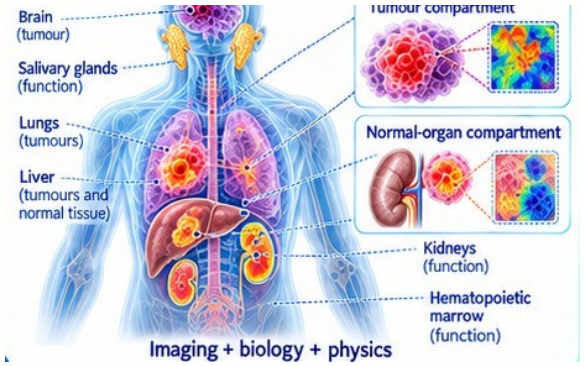
- Combine PET/CT, SPECT and MRI
- Burden, target expression, tracer kinetics
- Change over time; PSMA and SSTR theranostics





2 Characterize biology

- Heterogeneity within and between lesions
- Receptors, perfusion, hypoxia; normal-organ uptake and clearance
- Radiomics/AI + molecular and clinical data → stratification and targeting



3 Predict and personalize

- Digital twin of tumour + normal organs
- Predictive dosimetry: tumour control vs toxicity
- Virtual trials; models updated with new imaging

Goal: a predictive, therapy-guiding imaging platform — reproducible and clinically validated.

Concept adapted from Tharmaseelan et al., Cancers 2022;14:3349 (CC BY 4.0), extended to the proposed research vision

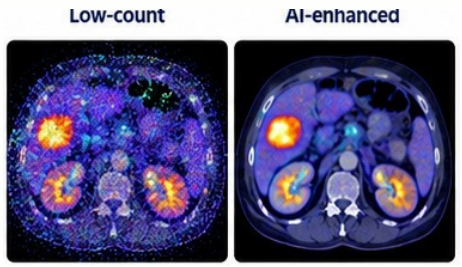
AIM II

AI-enhanced molecular imaging



1 Acquire

- Physics-informed reconstruction
- Lower-dose, faster PET/SPECT
- Quantitative accuracy kept



2 Enhance

- AI denoising
- Lesion detection and segmentation
- Registration over time



3 Quantify

- Disease burden and heterogeneity
- Radiomics + clinical + biology
- Personalized dosimetry



4 Support

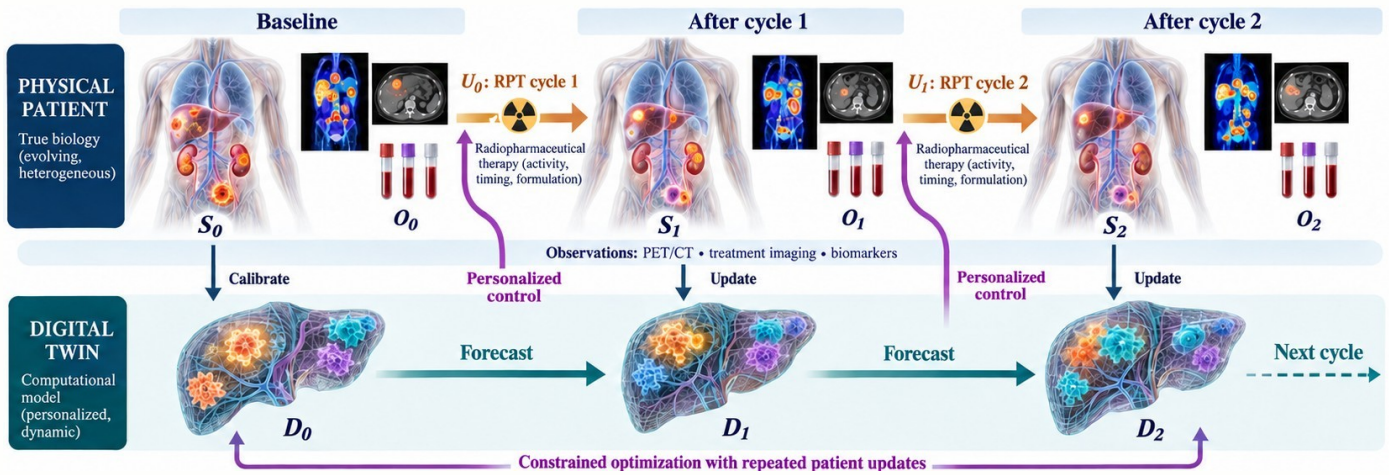
- Therapy-response prediction
- Patient stratification
- Interpretable decision support

Priorities: physical consistency · biological interpretability · robust validation — less repetitive work for clinicians.

Proposed research framework · informed by Herskovits (2021) and Mansouri et al. (2026)

AIM III

Digital twins for radiopharmaceutical therapy



Imaging and biomarkers calibrate the twin at baseline and update it after every cycle; the twin forecasts and sends back personalized control.

1 Mechanistic core

Pharmacokinetics → dose and dose rate → tumour growth, damage, repair, resistance

2 AI supports the model

Patient-specific parameters from imaging; fast surrogate simulations

3 Predict → optimize → treat

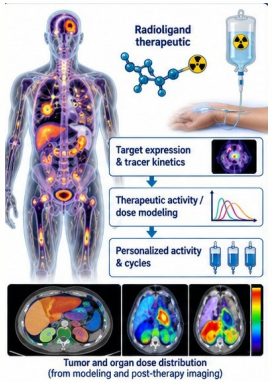
Predict response and toxicity → best activity, timing and number of cycles

Feedback loop: measure → predict → optimize → update — then validate prospectively.

Proposed RPT adaptation of Lorenzo et al., Annu Rev Biomed Eng 2024;26:529–560 (Figs. 4–6)

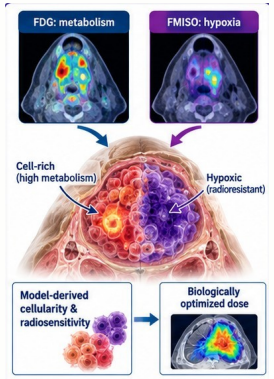
AIM IV

Advanced therapeutic technologies



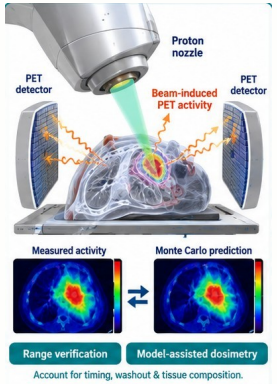
RPT planning

- PET target expression and kinetics
- Personalized activity and cycles
- Checked by post-therapy dosimetry



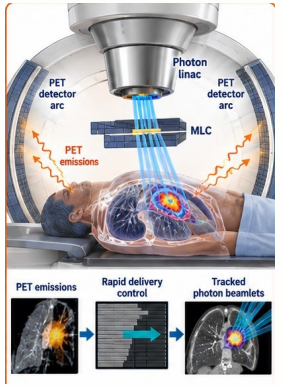
Dose painting

- FDG metabolism + FMISO hypoxia
- Map cellularity and radiosensitivity
- Respect normal-tissue limits



Proton therapy

- Beam-induced PET for range checks
- Model-assisted dosimetry
- PET activity ≠ absorbed dose



Biology-guided RT

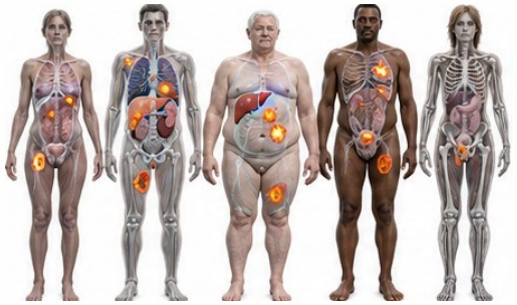
- Fast PET signals guide the beam
- Follows moving targets
- Within a planned dose envelope

Shared goal: plan, verify, measure response and adapt — including RPT + external-beam combinations.

Oderinde et al. (2021) · Lazzeroni et al. (2026) · Abdel-Wahab & Giammarile (2026)

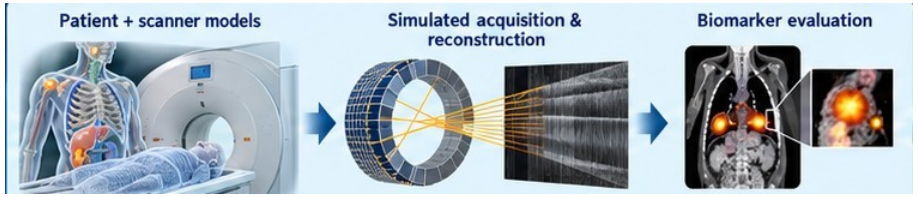
AIM V

Virtual trials and drug development

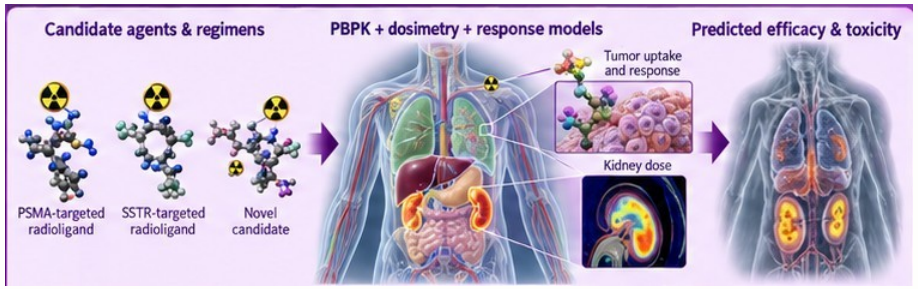


1 Virtual cohorts

- Imaging, biology and clinical data
- Anatomy, burden, kinetics, treatment sensitivity



2 Imaging trials: scanners, protocols, reconstruction; when is dynamic Ki better than static SUV?



3 Therapy trials: PBPK + dosimetry + response for agents, activities and schedules



4 Real-world studies

- Prioritize promising strategies
- Design more informative prospective trials

Credible simulation: physiological plausibility · uncertainty analysis · independent validation

Lindner & Link (2018) · Wangerin et al. (2017) · Kleeberger (2025) · Samei et al. (2025)